Incidence weights from a list with repeated representatives

The descent and transport results in the accompanying manuscript assume a complete list of distinct local conjugacy classes. A software interface may return a complete list with repetitions. The following elementary observation allows us to use that weaker input. It changes neither the descent theorem nor the classification being computed. We claim no new group-theoretic result.

Lemma 1. *Let H be a finite group. Suppose $M_1, \dots, M_s$ are maximal proper subgroups of H, and every maximal subgroup of H is H-conjugate to at least one M_i. Process the list in order, retaining M_i precisely when it is not H-conjugate to any previously retained subgroup. The retained list contains exactly one representative of each H-conjugacy class of maximal subgroups. Its completeness does not require the original list to be irredundant or to carry correct multiplicities.*

Proof. Two retained subgroups cannot be conjugate: the later one would have been discarded. For completeness, let M be any maximal subgroup and choose an index i for which M_i is conjugate to M. If M_i is retained, its class is represented. Otherwise the test that discards it supplies a previously retained representative of the same class. Thus every class is represented and no class is represented twice. All retained subgroups are maximal because they belong to the original list. □

If H is replaced by an isomorphic permutation group $\widetilde{H}$, this operation is performed using conjugacy in $\widetilde{H}$. The transport lemma carries the resulting distinct classes back to H. Conjugacy in a larger ambient group must not be used for this step: distinct local classes may fuse in that group, and each then contributes to the incidence number.

Lemma 2. *Let $H, K \leq G$ be finite groups. Among the retained representatives above, let $M_{i_1}, \dots, M_{i_t}$ be precisely those that are G-conjugate to K. Then the number of actual maximal subgroups of H in the ambient class $[K]_G$ is*

$$a_{H,K} = \sum_{j=1}^{t} [H : \mathrm{N}_H(M_{i_j})].$$

Writing $c_H = [G : \mathrm{N}_G(H)]$ and $c_K = [G : \mathrm{N}_G(K)]$, the reverse incidence is $b_{K,H} = c_H a_{H,K}/c_K$. Necessarily

$$0 \leq a_{H,K} \leq c_K, \qquad 0 \leq b_{K,H} \leq c_H.$$

In particular, if K is maximal in G, then $a_{G,K} = c_K$ and $b_{K,G} = 1$.

Proof. Conjugation by H has stabilizer $\mathrm{N}_H(M_{i_j})$ at M_{i_j}, so its orbit has the displayed index. The first lemma makes these orbits pairwise disjoint, and its completeness shows that their union is precisely the set being counted. Counting incident pairs of ambient conjugates first by their larger and then by their smaller subgroup gives the formula for $b_{K,H}$. Each fibre is a subset of the appropriate ambient conjugacy class, which proves both bounds. When the larger subgroup is G, all conjugates of a maximal K are maximal in it, and there is just one ambient subgroup conjugate to G. □

The implementation therefore discards repeated local conjugacy classes before ambient fusion and obtains each weight from a normalizer index. It does not use a multiplicity attached to the original list. The two bounds provide useful necessary checks; satisfying them alone does not establish correctness or completeness. Completeness of the original maximal-subgroup list and exactness of the group operations remain explicit computational premises.